

CLAUDINE HANDALI

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TECHNICAL SKILLS

Programming: Python, SQL, Typescript, Java, Javascript, HTML/CSS

Skills: Real-time telemetry, Time Series Analysis, Data Modeling, Dashboarding, Agentic AI, APIs

Tools: AWS, Git, DuckDB, Pandas, scikit-learn, D3.js, Mapbox GL JS, Flask, React, PyTorch, XGBoost

EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science (B.S.) in Data Science

Expected June 2026

Relevant Courseworks: Recommender Systems and Web mining, Scalable Systems and Analytics, Data Visualizations, Representation Learning, Data Analysis, Data Management, Data Structures and Algorithms, Machine Learning

WORK EXPERIENCE

Lead Student Researcher

January 2026 - Present

University of California, San Diego Center for Energy Research

La Jolla, CA

- Lead and mentor a team of 4 students in developing an interactive data dashboard analyzing the impact of EV adoption on oil import reduction across 25+ countries.
- Built quantitative models processing 10,000+ data points (energy consumption, trade flows, vehicle adoption rates) to estimate potential reductions in oil import expenditures.
- Presented findings to 10 faculty and researchers; dashboard adopted as a teaching and research tool within the center.

Student Researcher

June 2025 - October 2025

University of California, San Diego

La Jolla, CA

- Analyzed U.S. computing electricity consumption using 9TB of Intel DCA telemetry to profile CPU power-state residency and burst behavior, delivering plots and a technical brief that informed the team's next priorities.
- Built data processing pipelines and statistical models to evaluate computing efficiency across workloads.
- Standardized analysis notebooks using DuckDB on partitioned Parquet files.

AI Developer Intern

March 2025 - July 2025

Service Agent

La Jolla, CA

- Designed and deployed AI-driven automation agents, including a GPT-4-powered cold-calling system, using Python, Twilio, n8n, and Supabase.
- Developed data enrichment (web scraping + APIs) for CRM systems, improving data quality and completeness by 50%.
- Built predictive outreach algorithms integrating AI dialogue management, boosting conversion rates by 40%.

PROJECT EXPERIENCE

Predicting Recipe Popularity via Ratings

Python, Statsmodels, SciPy, NumPy, Multiple Linear Regression

- Analyzed 1.1M+ recipe interactions to identify key drivers of user ratings; found simpler recipes (≤ 8 ingredients) received 12–18% higher average ratings.
- Built a regression model to predict recipe popularity ($R^2 = 0.42$), quantifying the impact of complexity and prep time on user preferences.
- Improved model reliability through multicollinearity and heteroskedasticity checks.

End-to-End Analysis of U.S. Power Outage Durations

Python, Scikit-learn, Pandas, Seaborn, Random Forest Regression

- Modeled 100K+ outage events using Random Forest regression, reducing RMSE by 28% vs. linear baseline.
- Improved predictive performance from $R^2 = 0.51$ to 0.68 through hyperparameter tuning and feature engineering.
- Identified a 22% prediction error gap across outage types, highlighting bias and informing mitigation strategies

When It Rains: America's Changing Climate

JavaScript, D3.js, GeoJSON, Data Storytelling

- Processed and cleaned 170+ years (1850–2020) of NOAA precipitation data spanning 3,000+ countries and 60K+ data points.
- Built an interactive D3.js climate dashboard and emissions simulator, increasing user engagement 3× compared to static reports
- Designed intuitive visualizations enabling non-technical users to explore climate patterns without data expertise.